Chapter 33

The Nature and Propagation of Light

Learning Outcomes

In this chapter, you'll learn…

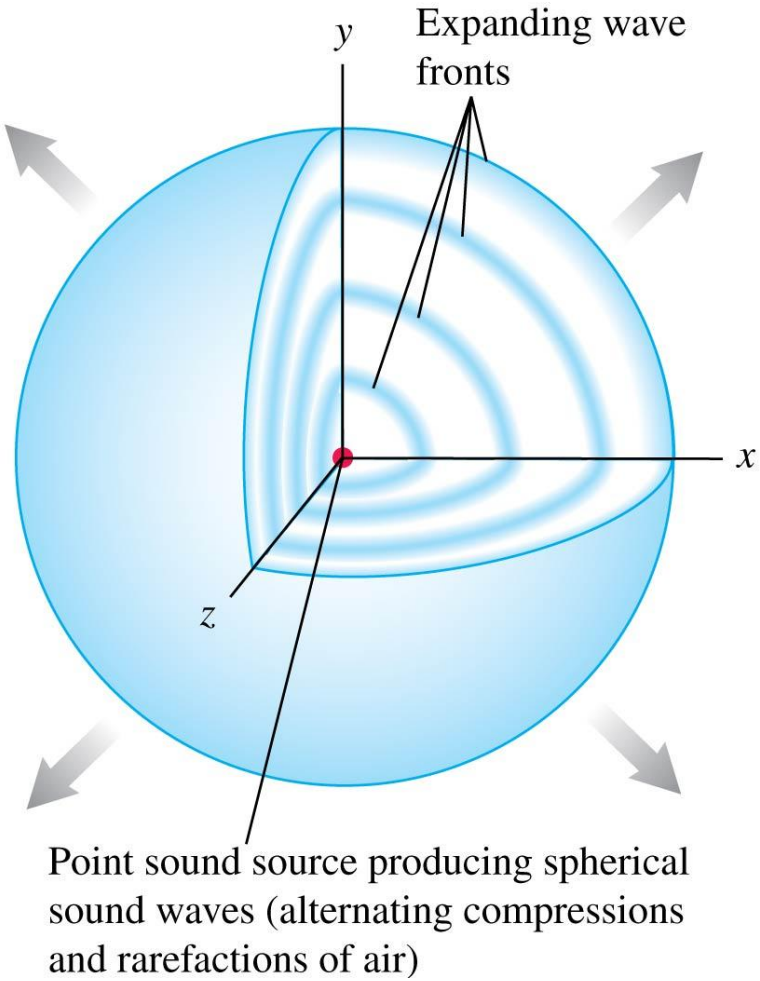
- what light rays are, and how they are related to wave fronts.
- the laws that govern the reflection and refraction of light.
- the circumstances under which light is totally reflected at an interface.
- how to make polarized light out of ordinary light.
- how the scattering of light explains the blue color of the sky.
- how Huygens's principle helps us analyze reflection and refraction.

Introduction

- When a cut diamond is illuminated with white light, it sparkles brilliantly with a spectrum of vivid colors.
- These distinctive visual features are a result of light traveling much slower in diamond than in air, and that light of different colors travels at different speeds in diamond.
- But by studying the branch of physics called **optics**, we can understand the blue color of the sky and the design of optical devices such as telescopes, microscopes, cameras, eyeglasses, and the human eye.

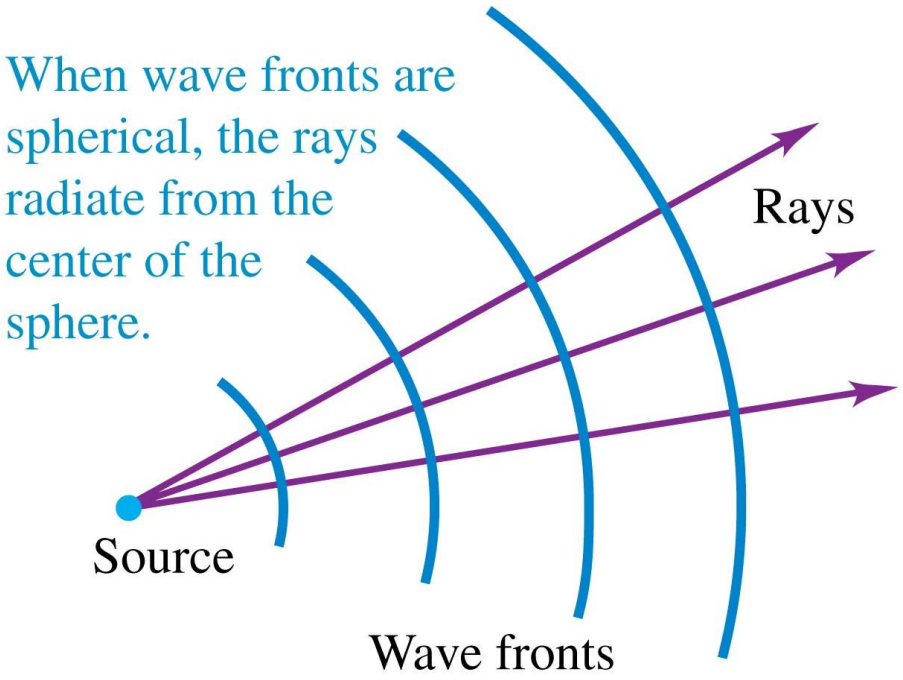
Waves and Wave Fronts

- A **wave front** is the locus of all adjacent points at which the **phase** of a wave is the same.
- Spherical wave fronts of sound spread out uniformly in all directions from a point source.
- Electromagnetic waves in vacuum also spread out as shown here.



Wave Fronts and Rays (1 of 2)

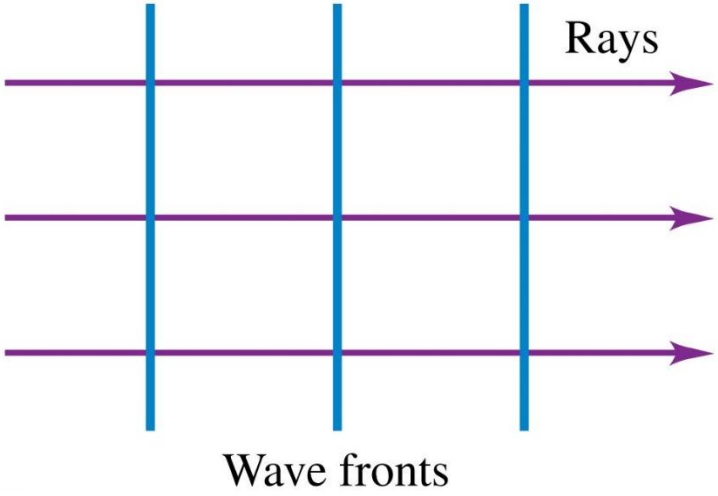
- It's often convenient to represent a light wave by **rays** rather than by wave fronts.
- A ray is an imaginary line along the direction of travel of the wave.
- When waves travel in a homogeneous isotropic material, the rays are always straight lines **normal** to the wave fronts.



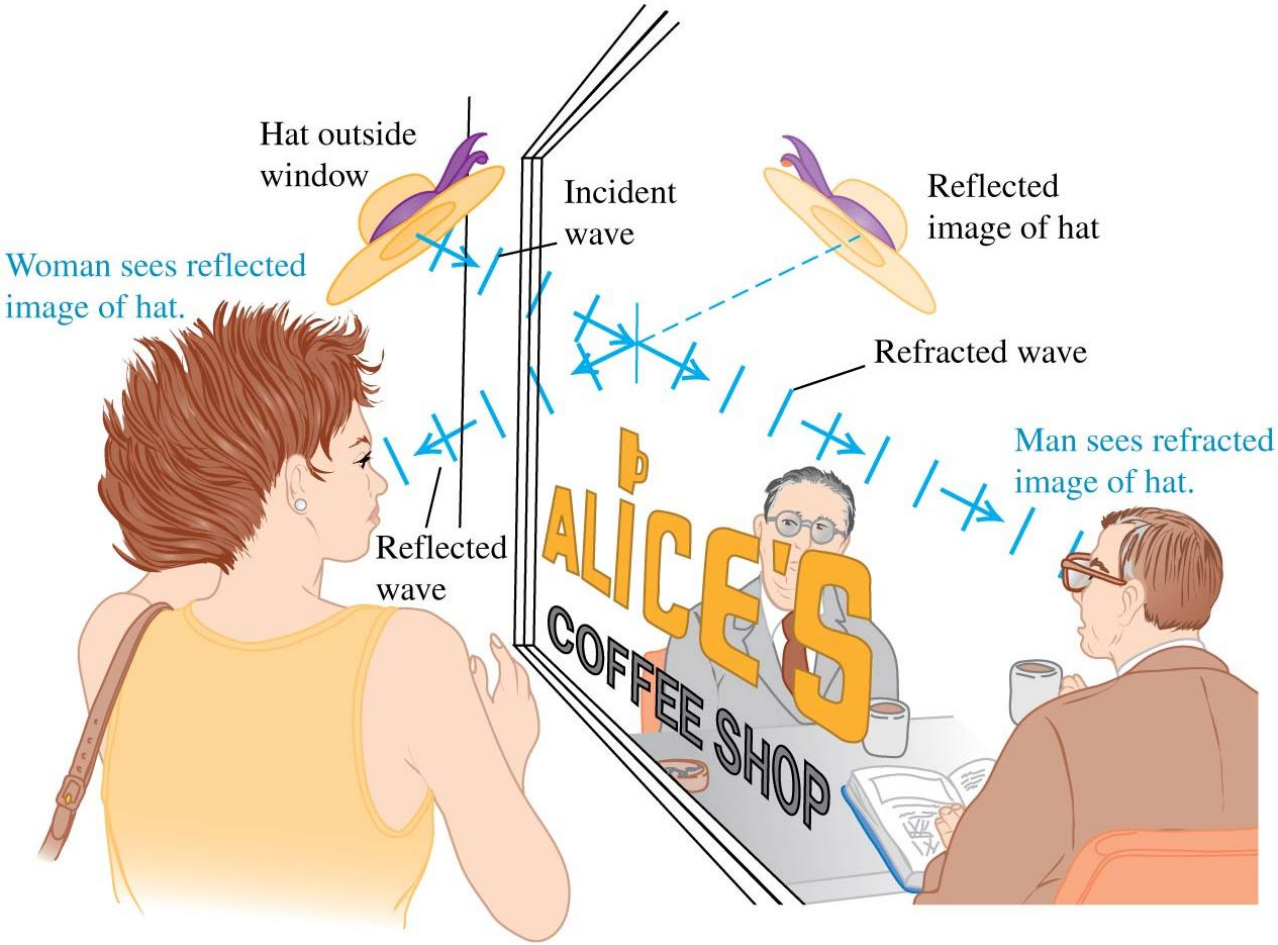
Wave Fronts and Rays (2 of 2)

- Far away from a source, where the radii of the spheres have become very large, a section of a spherical surface can be considered as a plane, and we have a **plane wave**.

When wave fronts are planar, the rays are perpendicular to the wave fronts and parallel to each other.

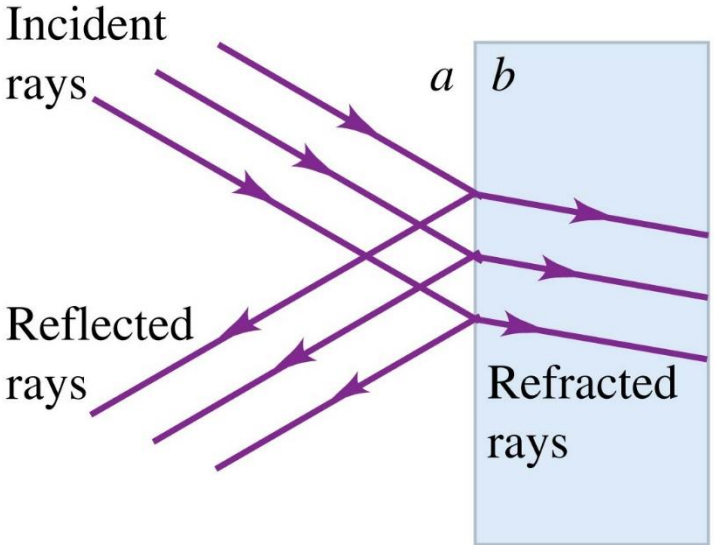


Reflection and Refraction (1 of 2)



Reflection and Refraction (2 of 2)

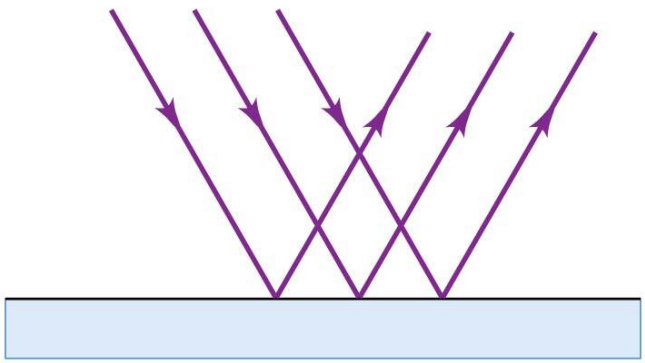
- When a light wave strikes a smooth interface separating two transparent materials (such as air and glass or water and glass), the wave is in general partly reflected and partly refracted (transmitted) into the second material.
- The segments of plane waves can be represented by bundles of rays forming beams of light.
- For simplicity we often draw only one ray in each beam.



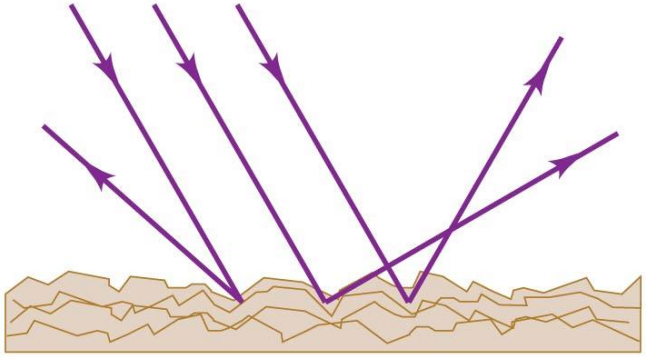
Diffuse and Specular Reflection

- Our primary concern in this chapter will be with **specular reflection** from a very smooth surface such as highly polished glass or metal (a).
- Scattered reflection from a rough surface is called **diffuse reflection** (b).
- The vast majority of objects in your environment are visible to you because they reflect light in a diffuse manner.

(a) Specular reflection



(b) Diffuse reflection



The Law of Reflection

- The angle of reflection is equal to the angle of incidence for all wavelengths and for any pair of materials.

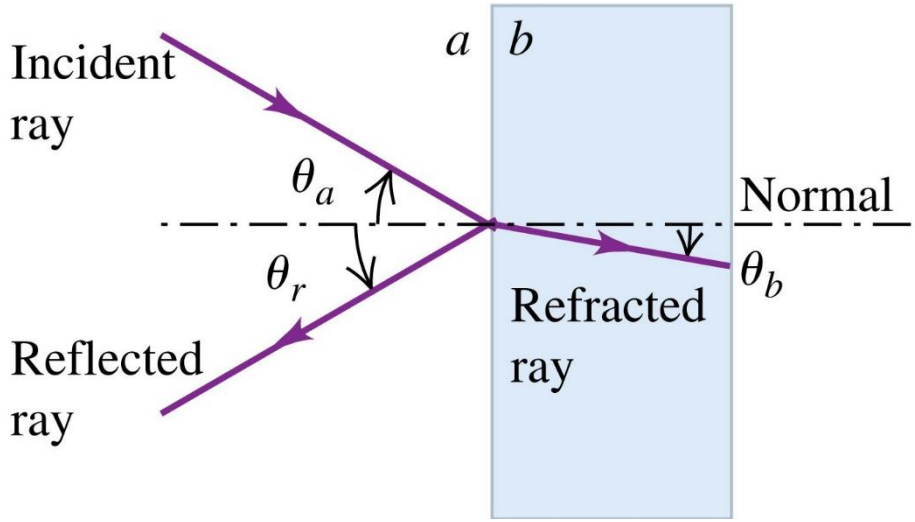
Law of reflection:

Angle of reflection (measured from normal)

$$\theta_r = \theta_a$$

Angle of incidence (measured from normal)

- Note that all angles are measured **from the normal**.



Index of Refraction

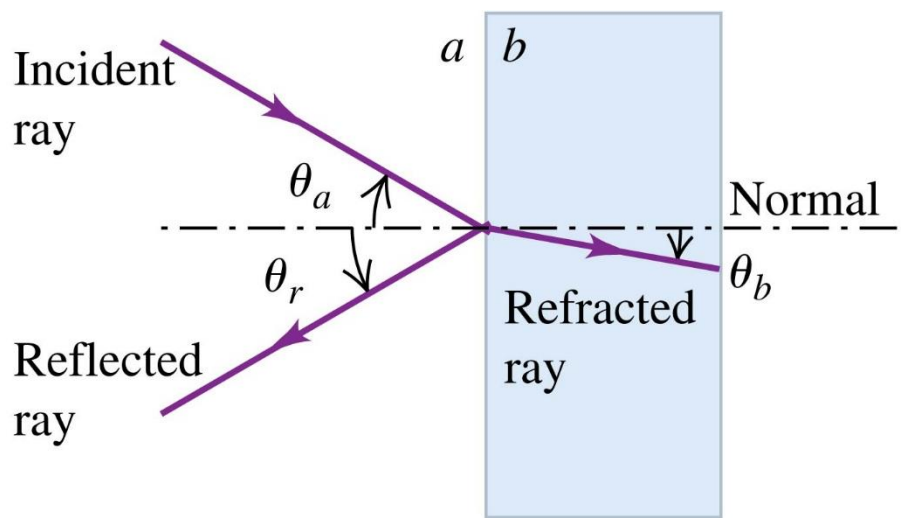
- The index of refraction of an optical material (also called the refractive index), denoted by *n*, is defined as:

Index of refraction of an optical material $\rightarrow$ $n = \dfrac{c}{v}$

c $\leftarrow$ Speed of light in vacuum

v $\leftarrow$ Speed of light in the material

- For the case shown here, material *b* has a **larger** index of refraction than material a ($n_b > n_a$) and the angle θ_b is **smaller** than θ_a.



The Law of Refraction

Law of refraction:

$$n_a \sin\theta_a = n_b \sin\theta_b$$

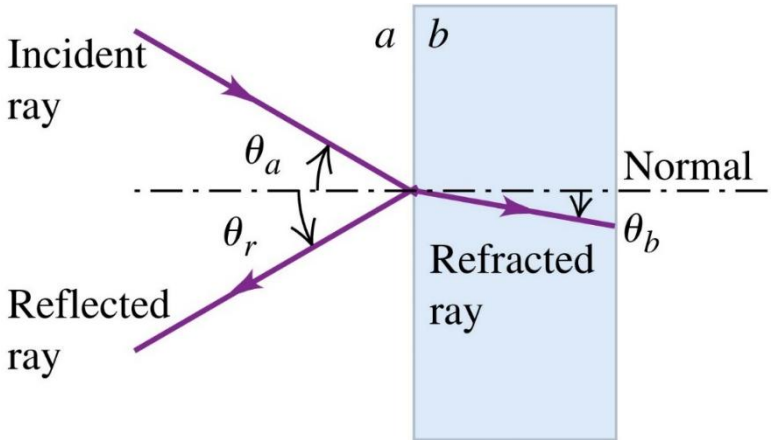
Angle of incidence (measured from normal)

Angle of refraction (measured from normal)

Index of refraction for material with incident light

Index of refraction for material with refracted light

- This result is also called Snell's law, after the Dutch scientist Willebrord Snell (1591–1626).

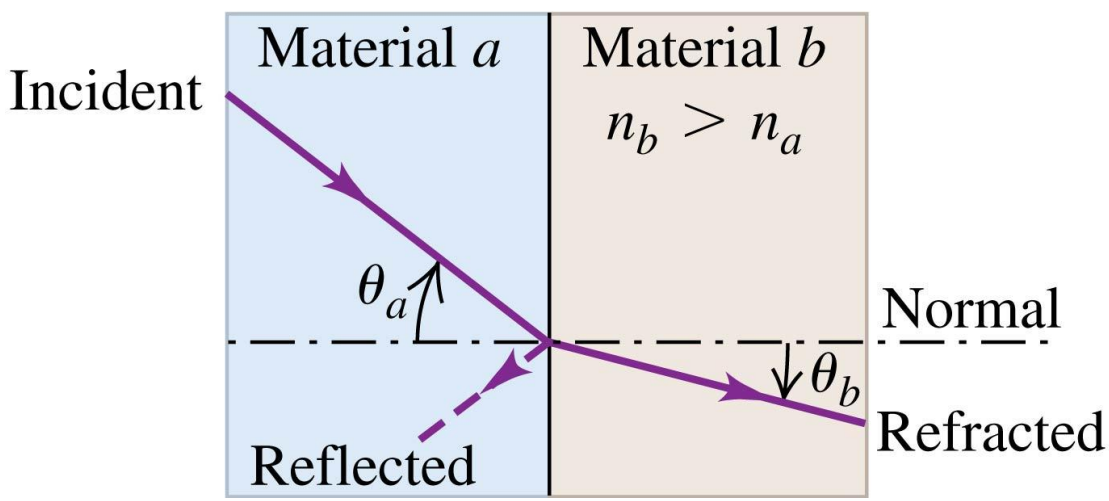


- Video Tutor Solution: Example 33.1

Reflection and Refraction: Case (1 of 3)

- When a ray passes from one material into another material having a **larger** index of refraction and hence a **slower** wave speed, the angle θ_b with the normal is smaller in the second material than the angle θ_a in the first.

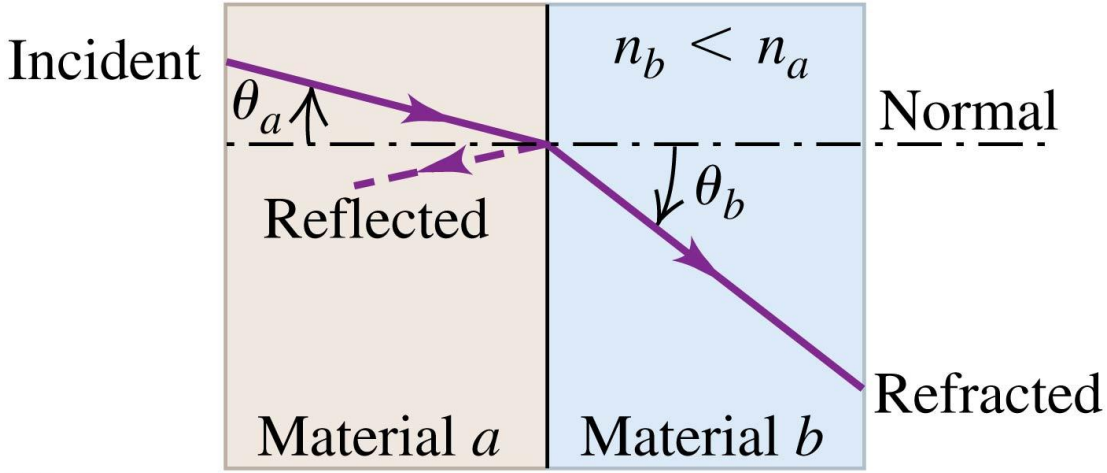
(a) A ray entering a material of *larger* index of refraction bends *toward* the normal.



Reflection and Refraction: Case (2 of 3)

- When a ray passes from one material into another material having a **smaller** index of refraction and hence a **faster** wave speed, the angle θ_b with the normal is **larger** in the second material than the angle θ_a in the first.

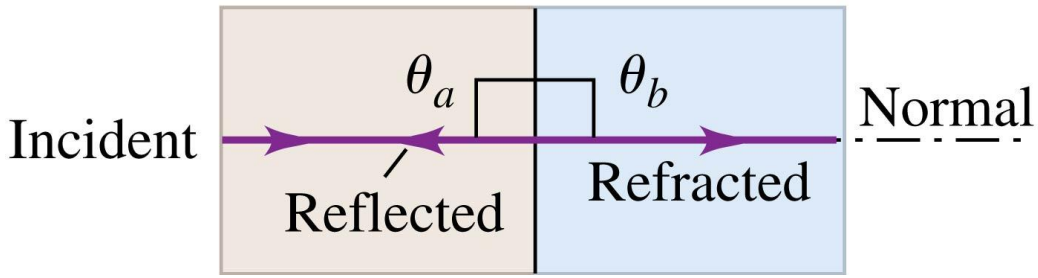
(b) A ray entering a material of *smaller* index of refraction bends *away from* the normal.



Reflection and Refraction: Case (3 of 3)

- In the case of normal incidence, the transmitted ray is not bent at all.
- In this case $\theta_a = 0$ and $\sin \theta_a = 0$, so θ_b is also equal to zero; the transmitted ray is also normal to the interface.
- θ_r is also equal to zero, so the reflected ray travels back along the same path as the incident ray.

(c) A ray oriented along the normal does not bend, regardless of the materials.



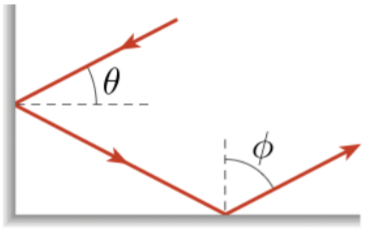
Why Does the Ruler Appear to Be Bent? (1 of 2)

- The law of refraction explains why a partially submerged straight ruler appears bent.
- Light rays coming from below the surface change in direction at the air–water interface, so the rays appear to be coming from a position above their actual point of origin.

Question 1

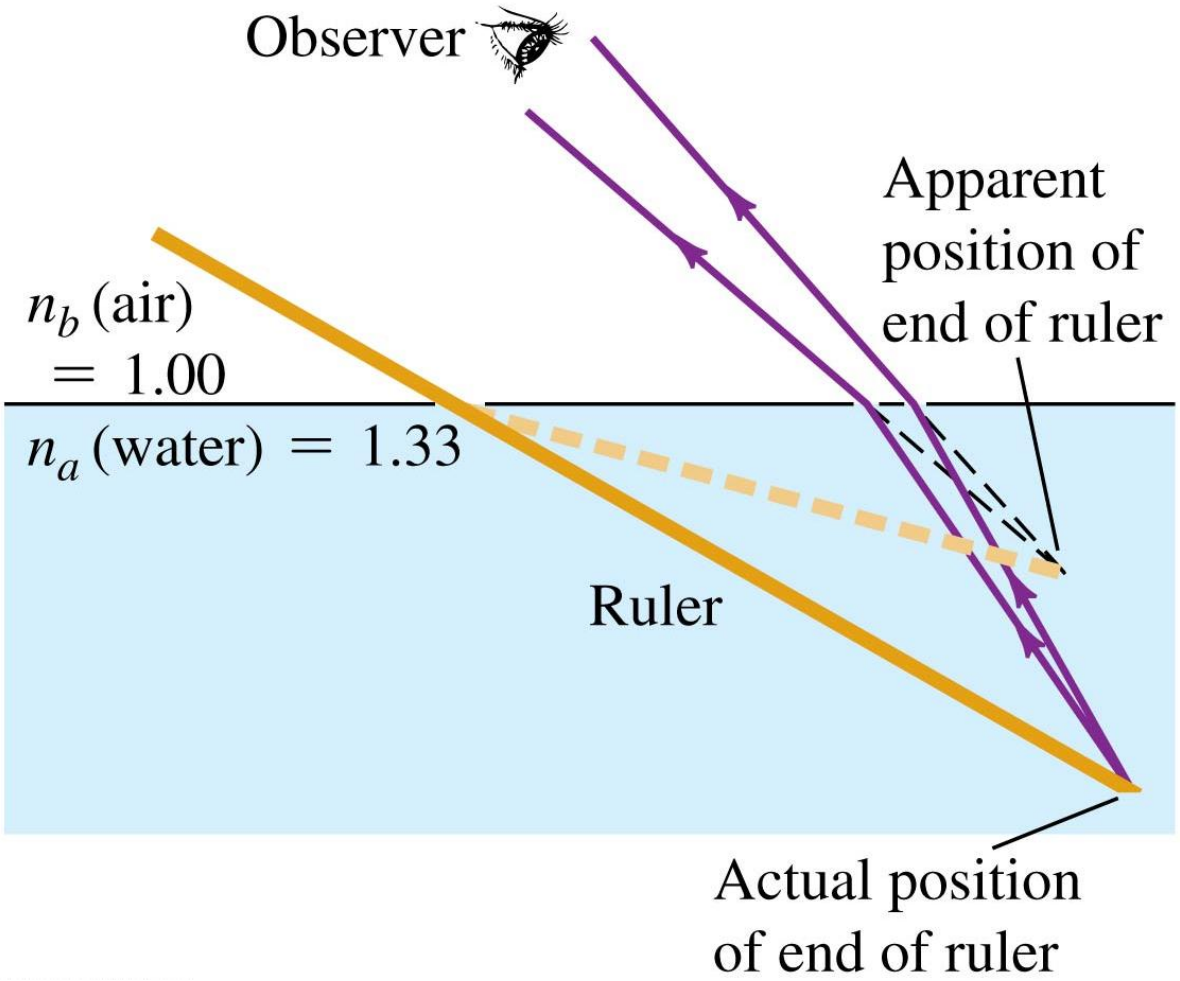
10 pts

Suppose two mirrors are arranged perpendicular to each other, and a ray of light is incident on the vertical mirror at an angle of 46° with respect to the normal, as in the figure. Find the angle ϕ with respect to the normal.



ϕ = _____ °

Why Does the Ruler Appear to Be Bent? (2 of 2)



Index of Refraction for Yellow Light (λ_0 = 589nm)

Substance	Index of Refraction, *n*
Ice (H_2O)	1.309
Water (H_2O) at 20 degrees Celsius	1.333
Glycerine at 20 degrees Celsius	1.473
Crown glass (typical value)	1.52
Rock salt (NaCl)	1.544
Quartz (SiO_2)	1.544
Diamond (C)	2.417

Index of Refraction and the Wave Aspects of Light

- The frequency *f* of a wave does not change when passing from one material to another.
- In any material, $v = \lambda f$; since *f* is the same in any material as in vacuum and *v* is always less than the wave speed *c* in vacuum, λ is also correspondingly reduced.

Wavelength of light in a material $\lambda = \dfrac{\lambda_0}{n}$

λ_0: Wavelength of light in vacuum

n: Index of refraction of the material

- When a wave passes from one material into a second material the waves get "squeezed" (the wavelength gets shorter) if the wave speed decreases and get "stretched" (the wavelength gets longer) if the wave speed increases.
- Video Tutor Solution: Example 33.2

Question 2

10 pts

Suppose the index of refraction of a certain medium is 3.39. Calculate the speed of light in that medium and supply the numerical prefix missing from the blank below.

$v =$ _____ $\times 10^8$ m/s

Question 3

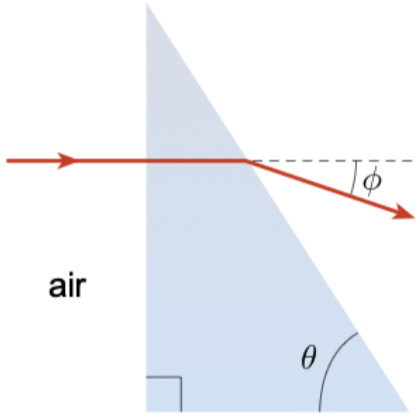
10 pts

The speed of light in a certain material is 121,567 km/s. Calculate its index of refraction.

Question 5

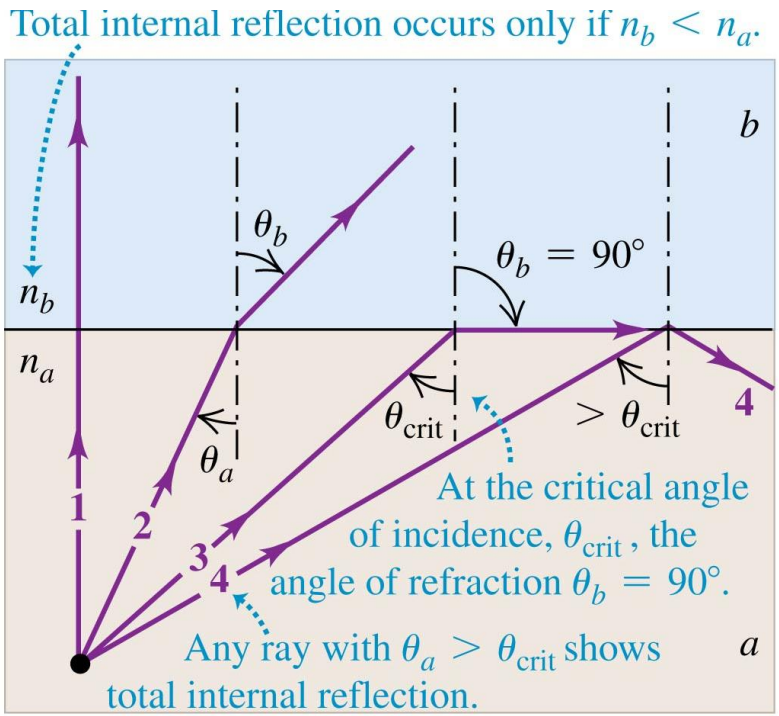
10 pts

The figure shows a light ray perpendicularly incident on a prism. Calculate the prism's index of refraction if θ = 53° and ϕ = 39°.



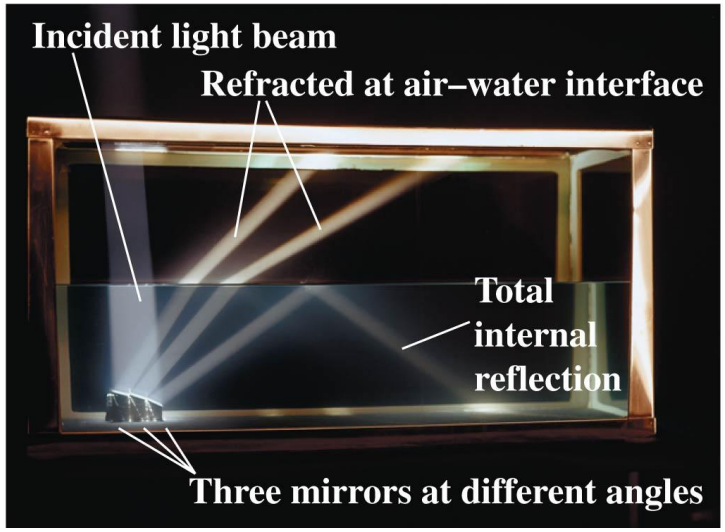
Total Internal Reflection (1 of 2)

- Under certain circumstances, **all** of the light can be reflected back from an interface, even though the second material is transparent.
- This is true for rays 3 and 4.
- The reflected portions of rays 1, 2, and 3 are omitted for clarity.



Total Internal Reflection (2 of 2)

- If the angle of incidence is larger than a **critical angle**, the ray cannot pass into the upper material; it is completely reflected at the boundary surface.
- This situation occurs only when $n_b < n_a$.



Critical angle for total internal reflection

$$\sin\theta_{\text{crit}} = \frac{n_b}{n_a}$$

n_b: Index of refraction of second material

n_a: Index of refraction of first material

Question 7

10 pts

A ray of light is traveling in a fluid having an index of refraction 1.67. It encounters a boundary with air and is totally internally reflected. Find the critical angle in degrees.

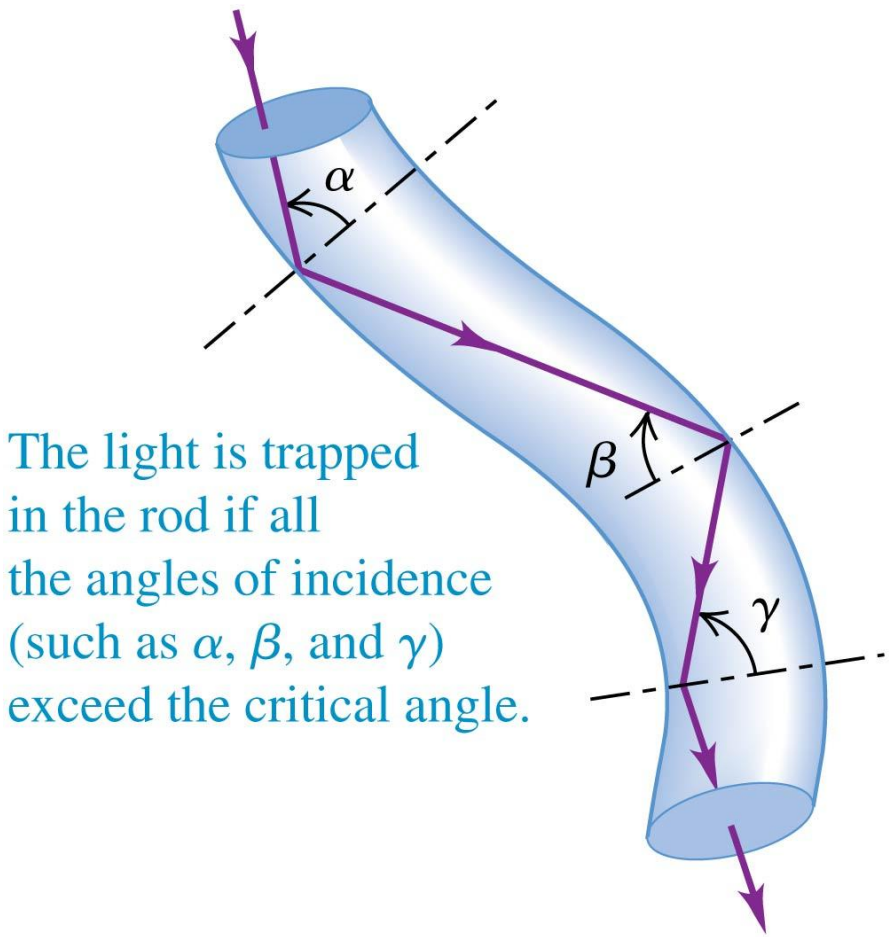
Question 8

10 pts

A liquid-air boundary has a critical angle of 41° for total internal reflection. If a ray of light traveling in air has an incident angle of 34° at the liquid-air boundary, calculate the angle between the refracted ray and the surface normal.

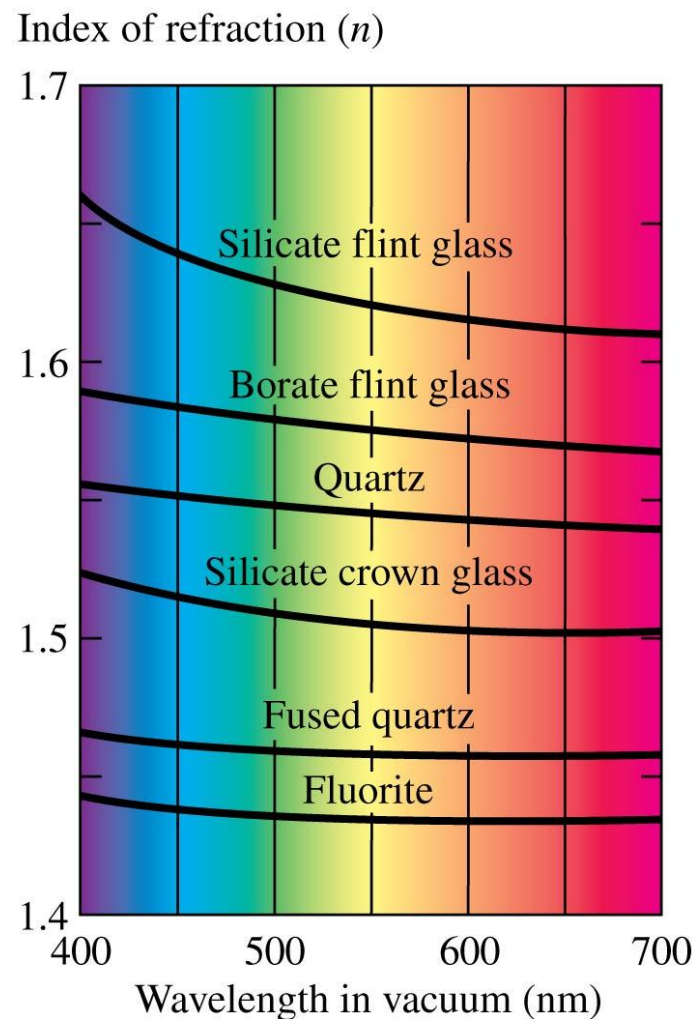
Fiber Optics

- When a beam of light enters at one end of a transparent rod, the light can be totally reflected internally if the index of refraction of the rod is greater than that of the surrounding material.
- The light is “trapped” within even a curved rod, provided that the curvature is not too great.



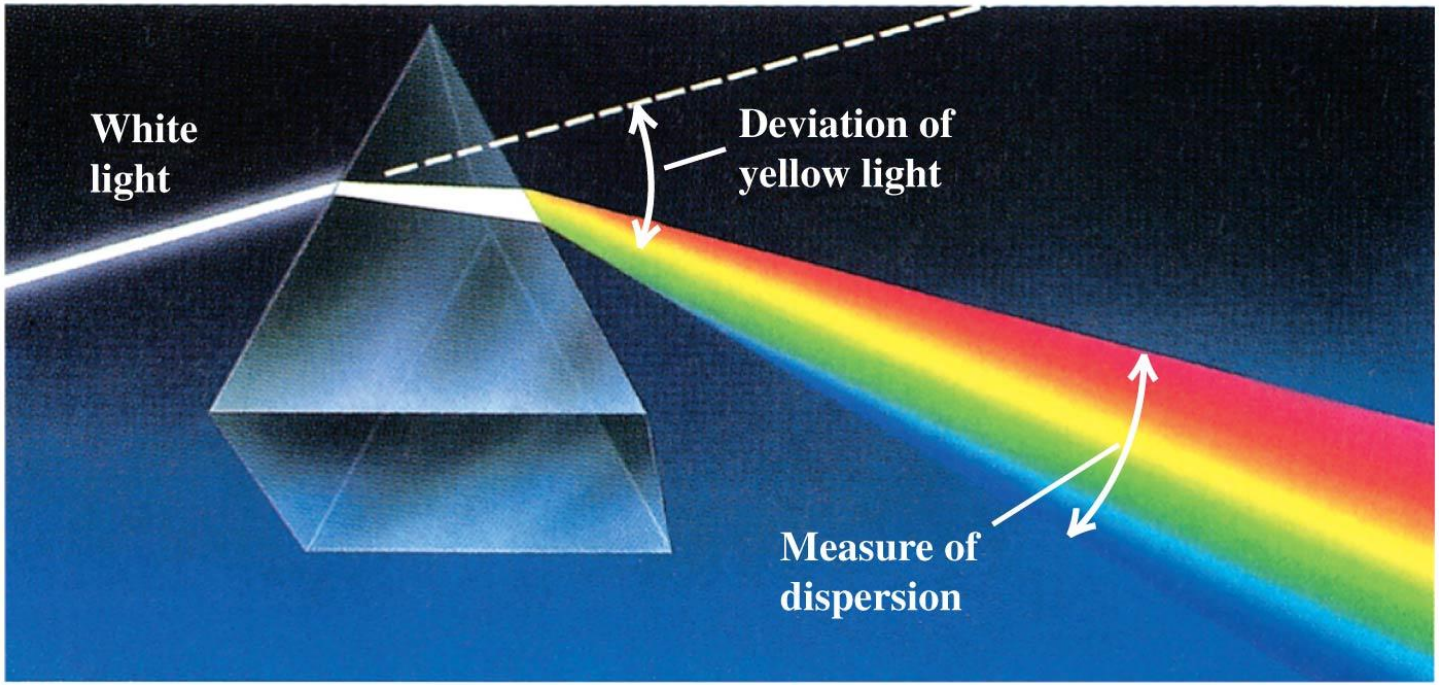
Dispersion (1 of 2)

- The speed of light in vacuum is the same for all wavelengths, but the speed in a material substance is different for different wavelengths.
- The dependence of wave speed and index of refraction on wavelength is called **dispersion**.
- In most materials the value of ***n*** **decreases** with increasing wavelength and decreasing frequency.



Dispersion (2 of 2)

- Ordinary white light is a superposition of waves with all visible wavelengths.
- The band of dispersed colors is called a **spectrum**.



Question 6

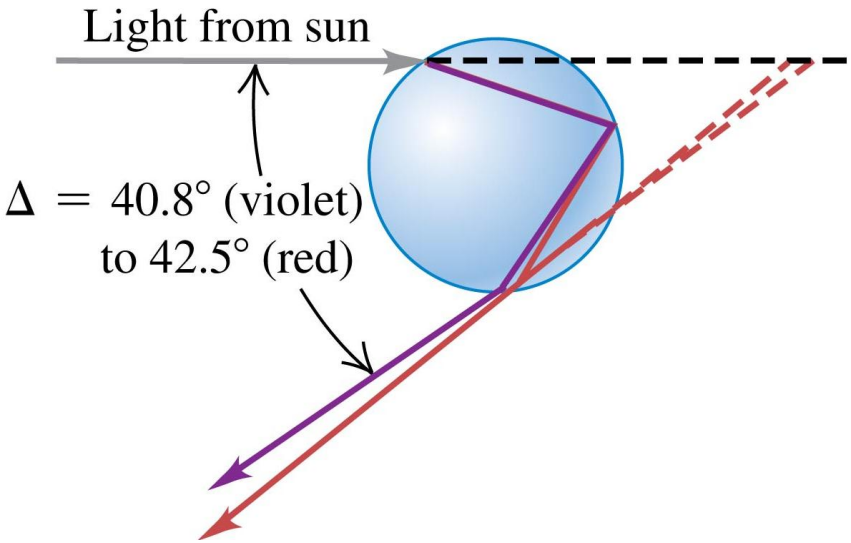
10 pts

The table below shows the index of refraction for BK7 optical glass at several different wavelengths. White light is incident on the top face of a large block of this glass at an angle of 51° with respect to the normal. If the block is 0.3 m thick, by how many millimeters are beams at wavelengths 2 and 4 separated at the bottom face of the block?

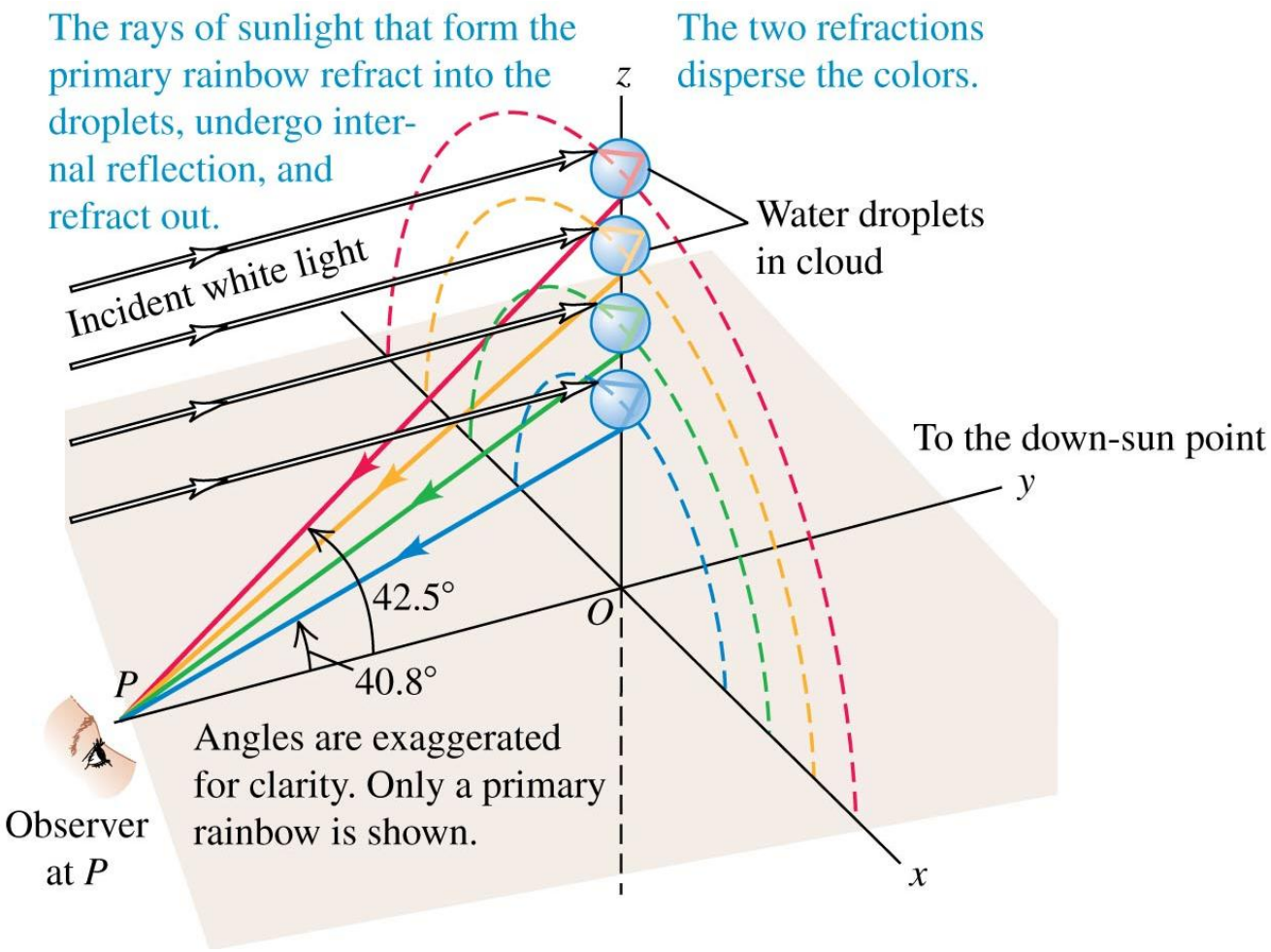
Number	Wavelength (nm)	Index of Refraction
1	400	1.531
2	475	1.523
3	550	1.519
4	625	1.515

How Rainbows Form (1 of 3)

- When sunlight enters a spherical water droplet suspended in the air, it is (partially) reflected from the back surface of the droplet, and is refracted again upon exiting the droplet.
- A light ray that enters the middle of the raindrop is reflected straight back.
- All other rays exit the raindrop within an angle Δ of that middle ray, with many rays “piling up” at the angle Δ.

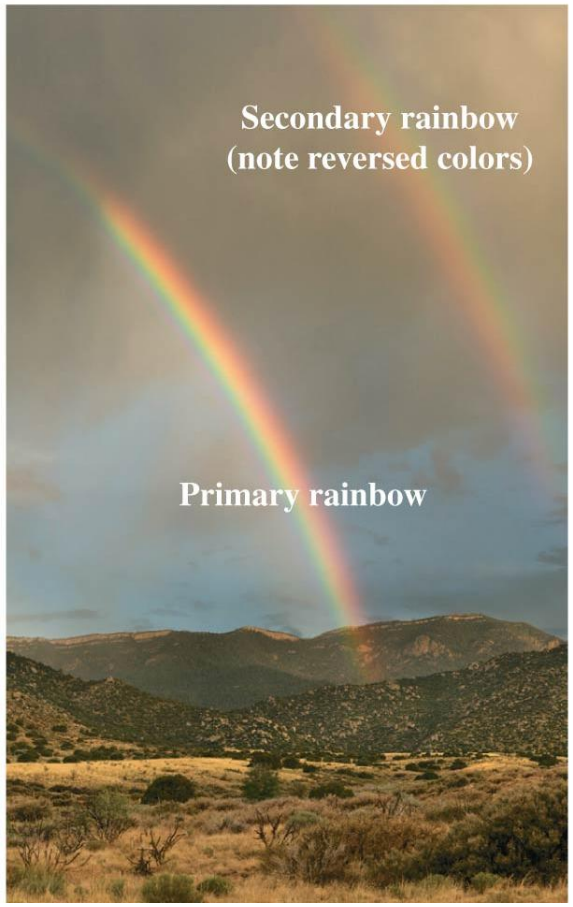


How Rainbows Form (2 of 3)



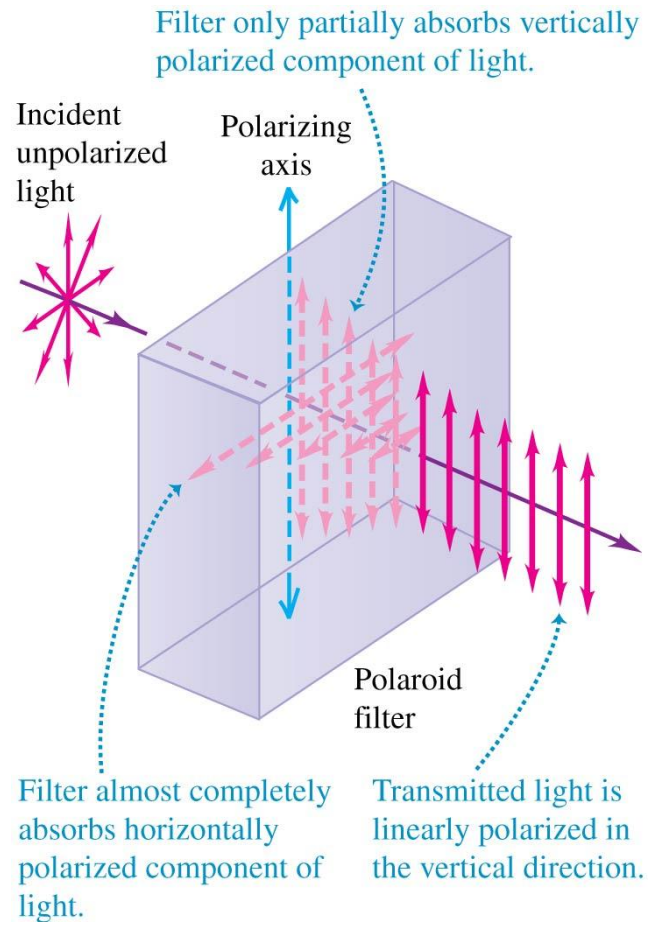
How Rainbows Form (3 of 3)

- In many cases you can see a second, larger rainbow.
- It is the result of two reflections from the back surface of the droplet.
- Just as a mirror held up to a book reverses the printed letters, so the second reflection reverses the sequence of colors in the secondary rainbow.



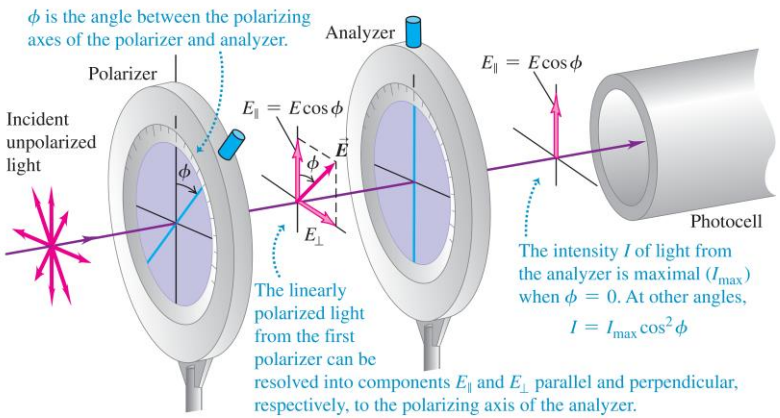
Polarization

- An electromagnetic wave is **linearly polarized** if the electric field has only one component.
- Light from most sources such as light bulbs is a random mixture of waves linearly polarized in all possible transverse directions; such light is called **unpolarized light** or **natural light**.
- A Polaroid **polarizing filter** can convert unpolarized light to linearly polarized light.
- Video Tutor Demonstration: Parallel-Wire Polarize for Microwaves



Malus's Law

- When polarized light of intensity I_{max} is incident on a polarizing filter used as an analyzer, the intensity I of the light transmitted through the analyzer depends on the angle ϕ between the polarization direction of the incident light and the polarizing axis of the analyzer.



Malus's law:

Intensity of polarized light passed through an analyzer

$$I = I_{max}\cos^2\phi$$

Angle between polarization axis of light and polarizing axis of analyzer

Maximum transmitted intensity

- Video Tutor Solution: Example 33.5

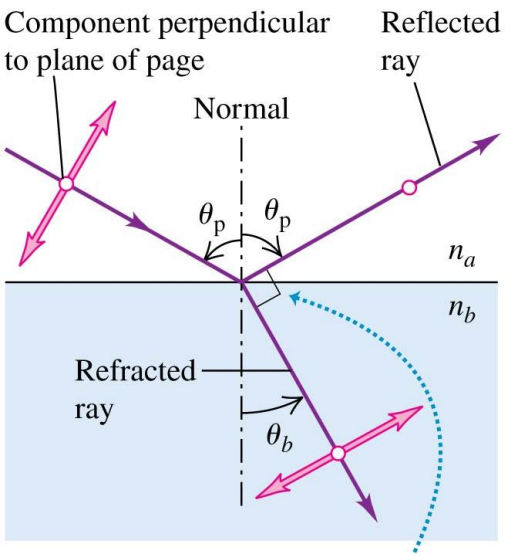
Question 9

10 pts

Unpolarized light of intensity 12 W/cm^2 is incident on a set of three polarizing filters, rotated 23°, 36°, and 23° from the vertical, respectively. Calculate the light intensity in W/cm^2 leaving the third filter.

Polarization by Reflection

- Unpolarized light can be polarized, either partially or totally, by reflection.
- At one particular angle of incidence, called the **polarizing angle**, the light for which $\vec{E}$ lies in the plane of incidence is not reflected at all but is completely refracted.



When light strikes a surface at the polarizing angle, the reflected and refracted rays are perpendicular to each other and

$$\tan\theta_p = \frac{n_b}{n_a}$$

Brewster's law for the polarizing angle:

$$\tan\theta_p = \frac{n_b}{n_a}$$

Polarizing angle (angle of incidence for which reflected light is 100% polarized)

n_b: Index of refraction of second material

n_a: Index of refraction of first material

Question 10

10 pts

Light traveling in air is incident on a fluid with index of refraction 2.16. At what angle of incidence with respect to the normal is the reflected light completely polarized? Answer in degrees.

Circular Polarization

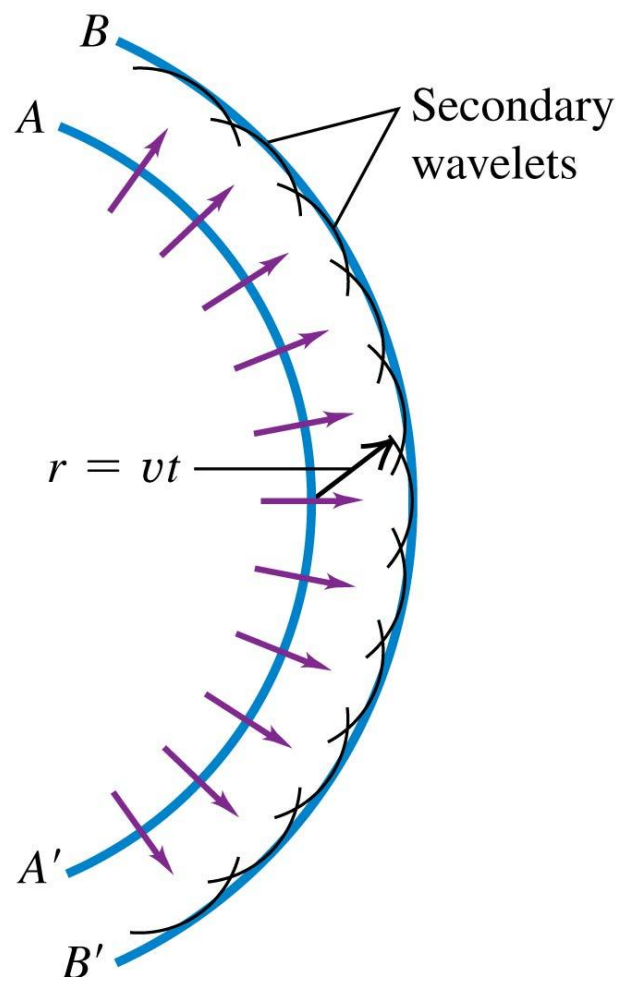
- **Circular polarization** occurs when the $\vec{E}$ vector has a constant magnitude but rotates around the direction of propagation.
- When the wave is propagating toward you and the $\vec{E}$ vector appears to be rotating clockwise, it is called a **right circularly polarized** electromagnetic wave.
- If instead the $\vec{E}$ vector of a wave coming toward you appears to be rotating counterclockwise, it is called a **left circularly polarized** electromagnetic wave.
- The lenses of the special glasses you wear to see a 3-D movie are circular polarizing filters.

Scattering of Light

- When you look at the daytime sky, the light that you see is sunlight that has been absorbed and then re-radiated in a variety of directions.
- This process is called **scattering**.
- Light scattered by air molecules contains 15 times as much blue light as red, and that's why the sky is blue.
- Clouds contain a high concentration of suspended water droplets or ice crystals, which scatter light of all wavelengths equally, so the cloud looks white.

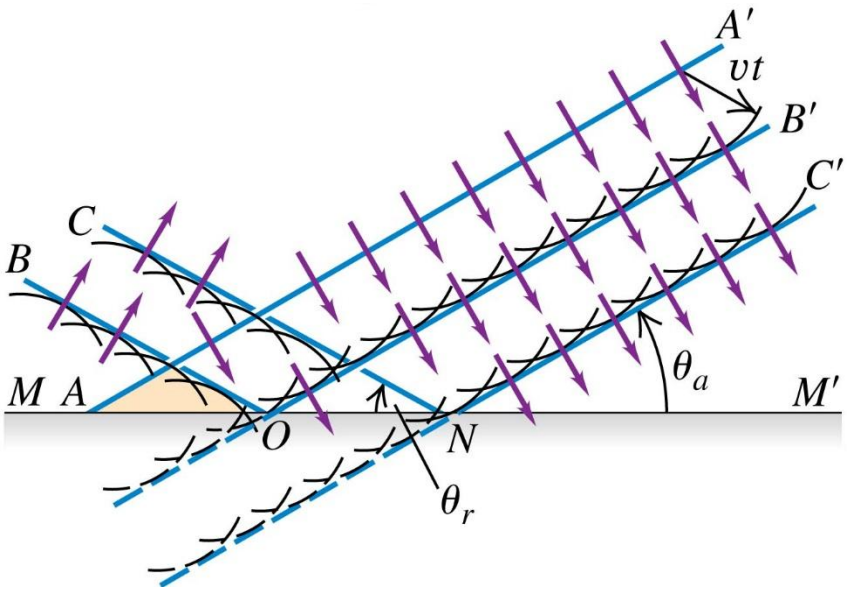
Huygens's Principle

- **Huygens's principle** states that **every point of a wave front may be considered the source of secondary wavelets that spread out in all directions with a speed equal to the speed of propagation of the wave**.
- The new wave front at a later time is then found by constructing a surface **tangent** to the secondary wavelets or, as it is called, the **envelope** of the wavelets.
- The figure shows the application of Huygens's principle to wave front *AA'* to construct a new wave front *BB'.*



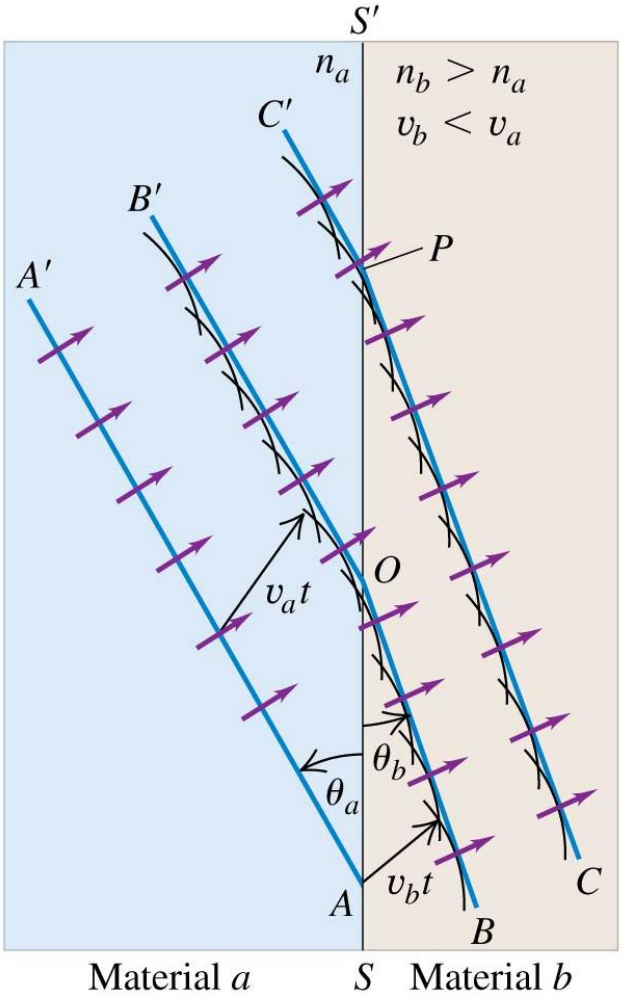
Reflection and Huygens's Principle

- To derive the law of reflection from Huygens's principle, we consider a plane wave approaching a plane reflecting surface.
- The effect of the reflecting surface is to change the direction of travel of those wavelets that strike it.
- The angle θ_a therefore equals the angle θ_r, and we have the law of reflection.



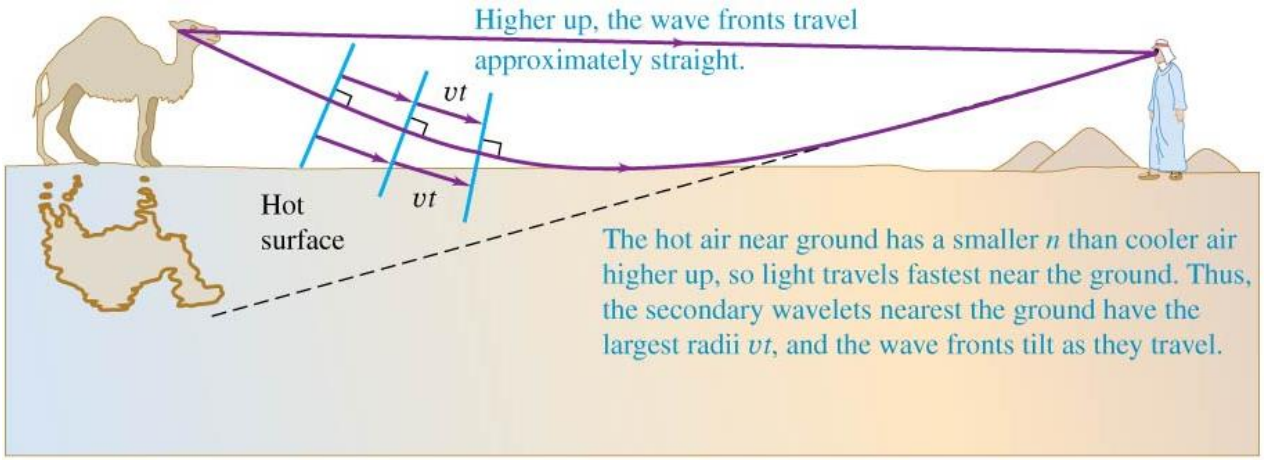
Refraction and Huygens's Principle

- Huygens's principle can be used to explain the law of refraction.
- Consider wave fronts traveling across the boundary surface *SS′* between two transparent materials *a* and *b*, with wave speeds $v_b < v_a$.
- We can apply Huygens's principle to find the relation of the angle θ_b to θ_a.



A Mirage

- Mirages are an example of Huygens's principle.
- A thirsty traveler can interpret the apparent reflecting surface as a sheet of water.



Question 11

10 pts

Total internal reflection can only occur when a ray of light encounters a medium with a [Select] index of refraction. For this reason, a ray traveling in a medium with index of refraction [Select] cannot undergo total internal reflection when encountering the boundary of another medium.

Question 12

10 pts

The larger index of refraction of a given material, the [Select] the speed of light in the material. Further, for a given nonzero angle with respect to the normal, a larger index of refraction means the ray will bend [Select] from its original trajectory upon entering the material from air.